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Course Catalog — B.Tech IT and M.Tech IT Courses

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Source-type legend used in tables: **O** = official (quoted from a GUNI PDF), **DO** = derived_official (structured/transformed from a GUNI PDF), **S** = synthetic (invented for CampusMate testing). Official page numbers refer to physical PDF pages of DOC-OFFICIAL-SYLLABUS-001 (GUNI structured syllabus) or DOC-OFFICIAL-CATALOG-001 (GUNI 1.1.3 course list).

Course-level records for every course code appearing in the B.Tech IT and M.Tech IT curricula of the official GUNI syllabus, plus five legacy IT course codes that appear only in the official certified course list. Each entry separates official fields [O], derived fields [DO] and synthetic enrichment [S].

A. B.Tech IT department courses

COURSE-2CEIT301 — 2CEIT301 DIGITAL ELECTRONICS

course_id	COURSE-2CEIT301
course_code [O]	2CEIT301
course_name [O]	DIGITAL ELECTRONICS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-03 (BTECH IT Semester III) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.12 — B.TECH-IT Semester III – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-016 Prof. Aman Circuitkar
laboratory [S]	LAB-IT-007 IoT & Robotics Lab (Room DB-101)
synthetic description [S]	Number systems, Boolean algebra, combinational and sequential logic design with hands-on trainer kits.

COURSE-2CEIT302 — 2CEIT302 OBJECT ORIENTED PROGRAMMING

course_id	COURSE-2CEIT302
course_code [O]	2CEIT302
course_name [O]	OBJECT ORIENTED PROGRAMMING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-03 (BTECH IT Semester III) [O]
course_type	regular
credits	5.0 (TH 3.0 + PR/TW 2.0); 7.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.12 — B.TECH-IT Semester III – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2ES104 PROGRAMMING FOR PROBLEM SOLVING
faculty [S]	FAC-IT-012 Prof. Ishaan Stackani
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Classes, objects, inheritance, polymorphism and exception handling with practical programming in an OO language.

COURSE-2CEIT303 — 2CEIT303 DATABASE MANAGEMENT SYSTEM

course_id	COURSE-2CEIT303
course_code [O]	2CEIT303
course_name [O]	DATABASE MANAGEMENT SYSTEM
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-03 (BTECH IT Semester III) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.12 — B.TECH-IT Semester III – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2ES104 PROGRAMMING FOR PROBLEM SOLVING
faculty [S]	FAC-IT-001 Dr. Meera Datavala, FAC-IT-005 Prof. Riya Queryani
laboratory [S]	LAB-IT-003 Database Systems Lab (Room DA-303)
synthetic description [S]	Relational model, ER modelling, SQL, normalisation, transactions and indexing, with practicals on PostgreSQL/MySQL.

COURSE-2CEIT304 — 2CEIT304 DATA STRUCTURES

course_id	COURSE-2CEIT304
course_code [O]	2CEIT304
course_name [O]	DATA STRUCTURES
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-03 (BTECH IT Semester III) [O]
course_type	regular
credits	5.0 (TH 3.0 + PR/TW 2.0); 7.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.12 — B.TECH-IT Semester III – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2ES104 PROGRAMMING FOR PROBLEM SOLVING
faculty [S]	FAC-IT-012 Prof. Ishaan Stackani
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)

synthetic description [S]	Arrays, stacks, queues, linked lists, trees, graphs, hashing, sorting and searching.

COURSE-2CEIT401 — 2CEIT401 OPERATING SYSTEMS

course_id	COURSE-2CEIT401
course_code [O]	2CEIT401
course_name [O]	OPERATING SYSTEMS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-04 (BTECH IT Semester IV) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.13 — B.TECH-IT Semester IV – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT304 DATA STRUCTURES
faculty [S]	FAC-IT-014 Prof. Rohan Kernelwala
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Processes, threads, CPU scheduling, synchronisation, memory management and file systems.

COURSE-2CEIT402 — 2CEIT402 DESIGN AND ANALYSIS OF ALGORITHMS

course_id	COURSE-2CEIT402
course_code [O]	2CEIT402
course_name [O]	DESIGN AND ANALYSIS OF ALGORITHMS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-04 (BTECH IT Semester IV) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.13 — B.TECH-IT Semester IV – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT304 DATA STRUCTURES
faculty [S]	FAC-IT-013 Dr. Pooja Lambdani
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Asymptotic analysis, divide and conquer, greedy, dynamic programming, graph algorithms and NP-completeness.

COURSE-2CEIT403 — 2CEIT403 APPLICATION DEVELOPMENT TOOLS

course_id	COURSE-2CEIT403
course_code [O]	2CEIT403
course_name [O]	APPLICATION DEVELOPMENT TOOLS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-04 (BTECH IT Semester IV) [O]
course_type	regular
credits	4.0 (TH 2.0 + PR/TW 2.0); 6.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.13 — B.TECH-IT Semester IV – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-015 Prof. Kavya Graphwala
laboratory [S]	LAB-IT-008 Web & Mobile Development Lab (Room DA-403)
synthetic description [S]	Modern application development toolchains, version control, build tools and rapid application frameworks.

COURSE-2CEIT404 — 2CEIT404 PROGRAMMING WITH PYTHON

course_id	COURSE-2CEIT404
course_code [O]	2CEIT404
course_name [O]	PROGRAMMING WITH PYTHON
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-04 (BTECH IT Semester IV) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.13 — B.TECH-IT Semester IV – course table
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-019 Dr. Farhan Pythonwala
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Python programming, data structures, modules, file handling and scientific libraries.

COURSE-2CEIT405 — 2CEIT405 NOSQL DATABASE SYSTEMS

course_id	COURSE-2CEIT405
course_code [O]	2CEIT405
course_name [O]	NOSQL DATABASE SYSTEMS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-04 (BTECH IT Semester IV) [O]
course_type	regular
credits	1.0 (TH 0.0 + PR/TW 1.0); 2.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.13 — B.TECH-IT Semester IV – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT303 DATABASE MANAGEMENT SYSTEM
faculty [S]	FAC-IT-005 Prof. Riya Queryani
laboratory [S]	LAB-IT-003 Database Systems Lab (Room DA-303)

synthetic description [S]	Practical introduction to document, key-value, column-family and graph databases.

COURSE-2CEIT501 — 2CEIT501 COMPUTER ARCHITECTURE & ORGANIZATION

course_id	COURSE-2CEIT501
course_code [O]	2CEIT501
course_name [O]	COMPUTER ARCHITECTURE & ORGANIZATION
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT301 DIGITAL ELECTRONICS
faculty [S]	FAC-IT-014 Prof. Rohan Kernelwala
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Instruction sets, CPU organisation, pipelining, memory hierarchy and I/O.

COURSE-2CEIT502 — 2CEIT502 SOFTWARE ENGINEERING

course_id	COURSE-2CEIT502
course_code [O]	2CEIT502
course_name [O]	SOFTWARE ENGINEERING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-017 Dr. Neha Testani
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Software process models, requirements, design, testing and project management.

COURSE-2CEIT503 — 2CEIT503 COMPUTER NETWORKS

course_id	COURSE-2CEIT503
course_code [O]	2CEIT503
course_name [O]	COMPUTER NETWORKS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [O]
course_type	regular
credits	5.0 (TH 4.0 + PR/TW 1.0); 6.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT401 OPERATING SYSTEMS
faculty [S]	FAC-IT-006 Dr. Vikram Packetkar
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Network layers, TCP/IP, routing, switching, transport protocols and socket programming.

COURSE-2CEIT504 — 2CEIT504 CAPSTONE PROJECT-I

course_id	COURSE-2CEIT504
course_code [O]	2CEIT504
course_name [O]	CAPSTONE PROJECT-I
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [O]
course_type	regular
credits	1.0 (TH 0.0 + PR/TW 1.0); 2.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—
synthetic description [S]	First stage of the capstone project: problem identification, literature survey and project proposal.

COURSE-2CEIT5PE1 — 2CEIT5PE1 ADVANCED JAVA

course_id	COURSE-2CEIT5PE1
course_code [O]	2CEIT5PE1
course_name [O]	ADVANCED JAVA
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [DO]
course_type	elective — group EGRP-BTECH-IT-05-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-012 Prof. Ishaan Stackani
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Elective option in group ELECTIVE-I: a course on Advanced Java covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT5PE2 — 2CEIT5PE2 COMPUTER GRAPHICS & VISUALIZATION

course_id	COURSE-2CEIT5PE2
course_code [O]	2CEIT5PE2
course_name [O]	COMPUTER GRAPHICS & VISUALIZATION
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [DO]
course_type	elective — group EGRP-BTECH-IT-05-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-011 Prof. Tanya Pixelani
laboratory [S]	LAB-IT-009 Vision & Graphics Lab (Room DA-404)

synthetic description [S]	Elective option in group ELECTIVE-I: a course on Computer Graphics & Visualization covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT5PE3 — 2CEIT5PE3 QUANTUM COMPUTING

course_id	COURSE-2CEIT5PE3
course_code [O]	2CEIT5PE3
course_name [O]	QUANTUM COMPUTING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [DO]
course_type	elective — group EGRP-BTECH-IT-05-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-019 Dr. Farhan Pythonwala
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Elective option in group ELECTIVE-I: a course on Quantum Computing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT5PE4 — 2CEIT5PE4 SOFTWARE PACKAGES

course_id	COURSE-2CEIT5PE4
course_code [O]	2CEIT5PE4
course_name [O]	SOFTWARE PACKAGES
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [DO]
course_type	elective — group EGRP-BTECH-IT-05-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-017 Dr. Neha Testani
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Elective option in group ELECTIVE-I: a course on Software Packages covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT5PE5 — 2CEIT5PE5 MOBILE APPLICATION DEVELOPMENT

course_id	COURSE-2CEIT5PE5
course_code [O]	2CEIT5PE5
course_name [O]	MOBILE APPLICATION DEVELOPMENT
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [DO]
course_type	elective — group EGRP-BTECH-IT-05-ELECTIVE-II
credits	4.0 (TH 2.0 + PR/TW 2.0); 6.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-015 Prof. Kavya Graphwala
laboratory [S]	LAB-IT-008 Web & Mobile Development Lab (Room DA-403)
synthetic description [S]	Elective option in group ELECTIVE-II: a course on Mobile Application Development covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT5PE6 — 2CEIT5PE6 HUMAN COMPUTER INTERACTION

course_id	COURSE-2CEIT5PE6
course_code [O]	2CEIT5PE6
course_name [O]	HUMAN COMPUTER INTERACTION
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [DO]
course_type	elective — group EGRP-BTECH-IT-05-ELECTIVE-II
credits	4.0 (TH 2.0 + PR/TW 2.0); 6.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-018 Prof. Yash Uxwala
laboratory [S]	LAB-IT-008 Web & Mobile Development Lab (Room DA-403)
synthetic description [S]	Elective option in group ELECTIVE-II: a course on Human Computer Interaction covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT5PE7 — 2CEIT5PE7 INNOVATION & ENTREPRENEURSHIP

course_id	COURSE-2CEIT5PE7
course_code [O]	2CEIT5PE7
course_name [O]	INNOVATION & ENTREPRENEURSHIP
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [DO]
course_type	elective — group EGRP-BTECH-IT-05-ELECTIVE-II
credits	4.0 (TH 2.0 + PR/TW 2.0); 6.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-018 Prof. Yash Uxwala
laboratory [S]	—
synthetic description [S]	Elective option in group ELECTIVE-II: a course on Innovation & Entrepreneurship covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT5PE8 — 2CEIT5PE8 ADVANCED PROGRAMMING WITH .NET

course_id	COURSE-2CEIT5PE8
course_code [O]	2CEIT5PE8
course_name [O]	ADVANCED PROGRAMMING WITH .NET
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [DO]
course_type	elective — group EGRP-BTECH-IT-05-ELECTIVE-II
credits	4.0 (TH 2.0 + PR/TW 2.0); 6.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-012 Prof. Ishaan Stackani
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Elective option in group ELECTIVE-II: a course on Advanced Programming With .Net covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT601 — 2CEIT601 THEORY OF COMPUTATION

course_id	COURSE-2CEIT601
course_code [O]	2CEIT601
course_name [O]	THEORY OF COMPUTATION
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2BS3102 DISCRETE MATHEMATICS & PROBABILITY
faculty [S]	FAC-IT-013 Dr. Pooja Lambdani
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)

synthetic description [S]	Finite automata, regular languages, context-free grammars, pushdown automata and Turing machines.

COURSE-2CEIT602 — 2CEIT602 ARTIFICIAL INTELLIGENCE

course_id	COURSE-2CEIT602
course_code [O]	2CEIT602
course_name [O]	ARTIFICIAL INTELLIGENCE
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT402 DESIGN AND ANALYSIS OF ALGORITHMS, COURSE-2BS3102 DISCRETE MATHEMATICS & PROBABILITY
faculty [S]	FAC-IT-002 Dr. Arjun Tensorkar, FAC-IT-004 Dr. Kabir Modelkar
laboratory [S]	LAB-IT-001 AI & Machine Learning Lab (Room DA-301)
synthetic description [S]	Search, knowledge representation, reasoning, planning and an introduction to machine learning, with lab exercises in Python.

COURSE-2CEIT603 — 2CEIT603 CLOUD COMPUTING

course_id	COURSE-2CEIT603
course_code [O]	2CEIT603
course_name [O]	CLOUD COMPUTING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT503 COMPUTER NETWORKS
faculty [S]	FAC-IT-009 Dr. Aditi Cloudani
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Virtualisation, cloud service and deployment models, cloud storage and containers.

COURSE-2CEIT604 — 2CEIT604 USER EXPERIENCE DESIGN

course_id	COURSE-2CEIT604
course_code [O]	2CEIT604
course_name [O]	USER EXPERIENCE DESIGN
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [O]
course_type	regular
credits	1.0 (TH 0.0 + PR/TW 1.0); 2.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – course table
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-018 Prof. Yash Uxwala
laboratory [S]	LAB-IT-008 Web & Mobile Development Lab (Room DA-403)
synthetic description [S]	User research, personas, wireframing, prototyping and usability testing.

COURSE-2CEIT605 — 2CEIT605 CAPSTONE PROJECT-II

course_id	COURSE-2CEIT605
course_code [O]	2CEIT605
course_name [O]	CAPSTONE PROJECT-II
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [O]
course_type	regular
credits	1.0 (TH 0.0 + PR/TW 1.0); 2.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT504 CAPSTONE PROJECT-I
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—
synthetic description [S]	Second stage of the capstone project: design and partial implementation.

COURSE-2CEIT6PE1 — 2CEIT6PE1 WEB TECHNOLOGY

course_id	COURSE-2CEIT6PE1
course_code [O]	2CEIT6PE1
course_name [O]	WEB TECHNOLOGY
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-015 Prof. Kavya Graphwala
laboratory [S]	LAB-IT-008 Web & Mobile Development Lab (Room DA-403)
synthetic description [S]	Elective option in group ELECTIVE-III: a course on Web Technology covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT6PE2 — 2CEIT6PE2 SOFTWARE VALIDATION & TESTING

course_id	COURSE-2CEIT6PE2
course_code [O]	2CEIT6PE2
course_name [O]	SOFTWARE VALIDATION & TESTING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-017 Dr. Neha Testani
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Elective option in group ELECTIVE-III: a course on Software Validation & Testing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT6PE3 — 2CEIT6PE3 CRYPTOGRAPHY AND NETWORK SECURITY

course_id	COURSE-2CEIT6PE3
course_code [O]	2CEIT6PE3
course_name [O]	CRYPTOGRAPHY AND NETWORK SECURITY
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT503 COMPUTER NETWORKS
faculty [S]	FAC-IT-007 Dr. Sana Cipherwala
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-III: a course on Cryptography And Network Security covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT6PE4 — 2CEIT6PE4 DATA SCIENCES AND MACHINE LEARNING

course_id	COURSE-2CEIT6PE4
course_code [O]	2CEIT6PE4
course_name [O]	DATA SCIENCES AND MACHINE LEARNING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT404 PROGRAMMING WITH PYTHON
faculty [S]	FAC-IT-004 Dr. Kabir Modelkar
laboratory [S]	LAB-IT-001 AI & Machine Learning Lab (Room DA-301)

synthetic description [S]	Data wrangling, exploratory analysis, supervised and unsupervised learning, and model evaluation.

COURSE-2CEIT6PE5 — 2CEIT6PE5 GAME DESIGN & DEVELOPMENT

course_id	COURSE-2CEIT6PE5
course_code [O]	2CEIT6PE5
course_name [O]	GAME DESIGN & DEVELOPMENT
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-018 Prof. Yash Uxwala
laboratory [S]	LAB-IT-008 Web & Mobile Development Lab (Room DA-403)
synthetic description [S]	Elective option in group ELECTIVE-III: a course on Game Design & Development covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT6PE6 — 2CEIT6PE6 ROBOTICS

course_id	COURSE-2CEIT6PE6
course_code [O]	2CEIT6PE6
course_name [O]	ROBOTICS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-IV
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-016 Prof. Aman Circuitkar
laboratory [S]	LAB-IT-007 IoT & Robotics Lab (Room DB-101)
synthetic description [S]	Elective option in group ELECTIVE-IV: a course on Robotics covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT6PE7 — 2CEIT6PE7 ETHICAL HACKING

course_id	COURSE-2CEIT6PE7
course_code [O]	2CEIT6PE7
course_name [O]	ETHICAL HACKING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-IV
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT503 COMPUTER NETWORKS
faculty [S]	FAC-IT-008 Prof. Dev Firewallani
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-IV: a course on Ethical Hacking covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT6PE8 — 2CEIT6PE8 IT INFRASTRUCTURE & MANAGEMENT

course_id	COURSE-2CEIT6PE8
course_code [O]	2CEIT6PE8
course_name [O]	IT INFRASTRUCTURE & MANAGEMENT
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-IV
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Elective option in group ELECTIVE-IV: a course on It Infrastructure & Management covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT6PE9 — 2CEIT6PE9 INTERNET OF THINGS

course_id	COURSE-2CEIT6PE9
course_code [O]	2CEIT6PE9
course_name [O]	INTERNET OF THINGS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [DO]
course_type	elective — group EGRP-BTECH-IT-06-ELECTIVE-IV
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-016 Prof. Aman Circuitkar
laboratory [S]	LAB-IT-007 IoT & Robotics Lab (Room DB-101)
synthetic description [S]	Elective option in group ELECTIVE-IV: a course on Internet Of Things covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT701 — 2CEIT701 COMPILER DESIGN

course_id	COURSE-2CEIT701
course_code [O]	2CEIT701
course_name [O]	COMPILER DESIGN
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – course table
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT601 THEORY OF COMPUTATION
faculty [S]	FAC-IT-013 Dr. Pooja Lambdani
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Lexical analysis, parsing, semantic analysis, intermediate code generation and optimisation.

COURSE-2CEIT702 — 2CEIT702 BIG DATA ANALYTICS

course_id	COURSE-2CEIT702
course_code [O]	2CEIT702
course_name [O]	BIG DATA ANALYTICS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – course table
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT303 DATABASE MANAGEMENT SYSTEM, COURSE-2CEIT405 NOSQL DATABASE SYSTEMS
faculty [S]	FAC-IT-010 Dr. Harsh Clusterwala
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Hadoop, Spark, distributed storage and large-scale data processing pipelines.

COURSE-2CEIT703 — 2CEIT703 CAPSTONE PROJECT-III

course_id	COURSE-2CEIT703
course_code [O]	2CEIT703
course_name [O]	CAPSTONE PROJECT-III
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [O]
course_type	regular
credits	6.0 (TH 0.0 + PR/TW 6.0); 12.0 contact hours; total marks 100 (TH int/ext None/None, PR int/ext 60/40) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – course table
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT605 CAPSTONE PROJECT-II
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—
synthetic description [S]	Final stage of the capstone project: complete implementation, testing, report and presentation.

COURSE-2CEIT704 — 2CEIT704 SOFT SKILLS

course_id	COURSE-2CEIT704
course_code [O]	2CEIT704
course_name [O]	SOFT SKILLS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [O]
course_type	regular
credits	1.0 (TH 0.0 + PR/TW 1.0); 2.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – course table
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—
synthetic description [S]	Communication, teamwork, interview and presentation skills for placement readiness.

COURSE-2CEIT78PE1 — 2CEIT78PE1 DEEP LEARNING

course_id	COURSE-2CEIT78PE1
course_code [O]	2CEIT78PE1
course_name [O]	DEEP LEARNING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT602 ARTIFICIAL INTELLIGENCE
faculty [S]	FAC-IT-002 Dr. Arjun Tensorkar
laboratory [S]	LAB-IT-001 AI & Machine Learning Lab (Room DA-301)
synthetic description [S]	Neural networks, CNNs, RNNs, transformers and training practice on GPUs.

COURSE-2CEIT78PE10 — 2CEIT78PE10 FOG COMPUTING

course_id	COURSE-2CEIT78PE10
course_code [O]	2CEIT78PE10
course_name [O]	FOG COMPUTING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-009 Dr. Aditi Cloudani
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on Fog Computing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE11 — 2CEIT78PE11 FAULT TOLERANT SYSTEMS

course_id	COURSE-2CEIT78PE11
course_code [O]	2CEIT78PE11
course_name [O]	FAULT TOLERANT SYSTEMS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-010 Dr. Harsh Clusterwala
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on Fault Tolerant Systems covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE12 — 2CEIT78PE12 HIGH PERFORMANCE COMPUTING

course_id	COURSE-2CEIT78PE12
course_code [O]	2CEIT78PE12
course_name [O]	HIGH PERFORMANCE COMPUTING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-010 Dr. Harsh Clusterwala
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on High Performance Computing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE13 — 2CEIT78PE13 COMPUTER VISION

course_id	COURSE-2CEIT78PE13
course_code [O]	2CEIT78PE13
course_name [O]	COMPUTER VISION
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT602 ARTIFICIAL INTELLIGENCE
faculty [S]	FAC-IT-011 Prof. Tanya Pixelani
laboratory [S]	LAB-IT-009 Vision & Graphics Lab (Room DA-404)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on Computer Vision covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE2 — 2CEIT78PE2 BLOCKCHAIN TECHNOLOGIES

course_id	COURSE-2CEIT78PE2
course_code [O]	2CEIT78PE2
course_name [O]	BLOCKCHAIN TECHNOLOGIES
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-008 Prof. Dev Firewallani
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Blockchain Technologies covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE3 — 2CEIT78PE3 FUNDAMENTALS OF IMAGE PROCESSING

course_id	COURSE-2CEIT78PE3
course_code [O]	2CEIT78PE3
course_name [O]	FUNDAMENTALS OF IMAGE PROCESSING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-011 Prof. Tanya Pixelani
laboratory [S]	LAB-IT-009 Vision & Graphics Lab (Room DA-404)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Fundamentals Of Image Processing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE4 — 2CEIT78PE4 MOBILE COMPUTING

course_id	COURSE-2CEIT78PE4
course_code [O]	2CEIT78PE4
course_name [O]	MOBILE COMPUTING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-015 Prof. Kavya Graphwala
laboratory [S]	LAB-IT-008 Web & Mobile Development Lab (Room DA-403)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Mobile Computing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE5 — 2CEIT78PE5 SEARCH ENGINE OPTIMIZATION

course_id	COURSE-2CEIT78PE5
course_code [O]	2CEIT78PE5
course_name [O]	SEARCH ENGINE OPTIMIZATION
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-005 Prof. Riya Queryani
laboratory [S]	LAB-IT-002 Natural Language Processing Lab (Room DA-302)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Search Engine Optimization covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE6 — 2CEIT78PE6 VIRTUAL REALITY

course_id	COURSE-2CEIT78PE6
course_code [O]	2CEIT78PE6
course_name [O]	VIRTUAL REALITY
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-011 Prof. Tanya Pixelani
laboratory [S]	LAB-IT-009 Vision & Graphics Lab (Room DA-404)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Virtual Reality covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE7 — 2CEIT78PE7 NATURAL LANGUAGE PROCESSING

course_id	COURSE-2CEIT78PE7
course_code [O]	2CEIT78PE7
course_name [O]	NATURAL LANGUAGE PROCESSING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-2CEIT602 ARTIFICIAL INTELLIGENCE
faculty [S]	FAC-IT-003 Dr. Nisha Tokenwala
laboratory [S]	LAB-IT-002 Natural Language Processing Lab (Room DA-302)
synthetic description [S]	Text preprocessing, language models, embeddings, sequence labelling, transformers and retrieval-augmented generation.

COURSE-2CEIT78PE8 — 2CEIT78PE8 FORENSICS & CYBER LAW

course_id	COURSE-2CEIT78PE8
course_code [O]	2CEIT78PE8
course_name [O]	FORENSICS & CYBER LAW
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-008 Prof. Dev Firewallani
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on Forensics & Cyber Law covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT78PE9 — 2CEIT78PE9 ADVANCED ALGORITHMS

course_id	COURSE-2CEIT78PE9
course_code [O]	2CEIT78PE9
course_name [O]	ADVANCED ALGORITHMS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-07 (BTECH IT Semester VII) [DO]
course_type	elective — group EGRP-BTECH-IT-07-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.16 — B.TECH-IT Semester VII – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-013 Dr. Pooja Lambdani
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on Advanced Algorithms covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2CEIT801 — 2CEIT801 INDUSTRIAL TRAINING

course_id	COURSE-2CEIT801
course_code [O]	2CEIT801
course_name [O]	INDUSTRIAL TRAINING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-08 (BTECH IT Semester VIII) [O]
course_type	regular
credits	20.0 (TH 0.0 + PR/TW 20.0); 40.0 contact hours; total marks 400 (TH int/ext None/None, PR int/ext 200/200) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.18 — B.TECH-IT Semester VIII – course table
also in syllabus of [DO]	B.TECH-CE
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—
synthetic description [S]	Semester-long industrial training in a host organisation with periodic reporting.

B. M.Tech IT department courses

COURSE-3CEIT101 — 3CEIT101 COMPUTATIONAL INTELLIGENCE

course_id	COURSE-3CEIT101
course_code [O]	3CEIT101
course_name [O]	COMPUTATIONAL INTELLIGENCE
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-002 Dr. Arjun Tensorkar
laboratory [S]	LAB-IT-001 AI & Machine Learning Lab (Room DA-301)
synthetic description [S]	Fuzzy systems, evolutionary algorithms, neural networks and swarm intelligence.

COURSE-3CEIT102 — 3CEIT102 COMPUTER ALGORITHMS

course_id	COURSE-3CEIT102
course_code [O]	3CEIT102
course_name [O]	COMPUTER ALGORITHMS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-013 Dr. Pooja Lambdani
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	—

COURSE-3CEIT103 — 3CEIT103 CRYPTOGRAPHY AND NETWORK SECURITY

course_id	COURSE-3CEIT103
course_code [O]	3CEIT103
course_name [O]	CRYPTOGRAPHY AND NETWORK SECURITY
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-007 Dr. Sana Cipherwala
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	—

COURSE-3CEIT104 — 3CEIT104 RESEARCH ORIENTATION

course_id	COURSE-3CEIT104
course_code [O]	3CEIT104
course_name [O]	RESEARCH ORIENTATION
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [O]
course_type	regular
credits	1.0 (TH 0.0 + PR/TW 1.0); 2.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—
synthetic description [S]	—

COURSE-3CEIT105 — 3CEIT105 WIRELESS NETWORKS

course_id	COURSE-3CEIT105
course_code [O]	3CEIT105
course_name [O]	WIRELESS NETWORKS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-006 Dr. Vikram Packetkar
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-I: a course on Wireless Networks covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT106 — 3CEIT106 ADVANCED TOPICS IN NETWORKS

course_id	COURSE-3CEIT106
course_code [O]	3CEIT106
course_name [O]	ADVANCED TOPICS IN NETWORKS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-006 Dr. Vikram Packetkar
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-I: a course on Advanced Topics In Networks covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT107 — 3CEIT107 INTERNET TECHNOLOGIES

course_id	COURSE-3CEIT107
course_code [O]	3CEIT107
course_name [O]	INTERNET TECHNOLOGIES
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-015 Prof. Kavya Graphwala
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-I: a course on Internet Technologies covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT108 — 3CEIT108 NETWORK PROGRAMMING AND MANAGEMENT

course_id	COURSE-3CEIT108
course_code [O]	3CEIT108
course_name [O]	NETWORK PROGRAMMING AND MANAGEMENT
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-006 Dr. Vikram Packetkar
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-I: a course on Network Programming And Management covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT109 — 3CEIT109 HIGH PERFORMANCE COMPUTING

course_id	COURSE-3CEIT109
course_code [O]	3CEIT109
course_name [O]	HIGH PERFORMANCE COMPUTING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-I
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-010 Dr. Harsh Clusterwala
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Elective option in group ELECTIVE-I: a course on High Performance Computing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT110 — 3CEIT110 DISTRIBUTED COMPUTING

course_id	COURSE-3CEIT110
course_code [O]	3CEIT110
course_name [O]	DISTRIBUTED COMPUTING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-II
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-009 Dr. Aditi Cloudani
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Elective option in group ELECTIVE-II: a course on Distributed Computing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT111 — 3CEIT111 SERVICE ORIENTED COMPUTING

course_id	COURSE-3CEIT111
course_code [O]	3CEIT111
course_name [O]	SERVICE ORIENTED COMPUTING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-II
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-009 Dr. Aditi Cloudani
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Elective option in group ELECTIVE-II: a course on Service Oriented Computing covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT112 — 3CEIT112 ADVANCE SOFTWARE ENGINEERING

course_id	COURSE-3CEIT112
course_code [O]	3CEIT112
course_name [O]	ADVANCE SOFTWARE ENGINEERING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-II
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-017 Dr. Neha Testani
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Elective option in group ELECTIVE-II: a course on Advance Software Engineering covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT113 — 3CEIT113 REAL TIME SYSTEMS

course_id	COURSE-3CEIT113
course_code [O]	3CEIT113
course_name [O]	REAL TIME SYSTEMS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-01 (MTECH IT Semester I) [DO]
course_type	elective — group EGRP-MTECH-IT-01-ELECTIVE-II
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.104 — M.TECH-IT Semester I – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-014 Prof. Rohan Kernelwala
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Elective option in group ELECTIVE-II: a course on Real Time Systems covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT201 — 3CEIT201 DIGITAL IMAGE PROCESSING

course_id	COURSE-3CEIT201
course_code [O]	3CEIT201
course_name [O]	DIGITAL IMAGE PROCESSING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-011 Prof. Tanya Pixelani
laboratory [S]	LAB-IT-009 Vision & Graphics Lab (Room DA-404)
synthetic description [S]	—

COURSE-3CEIT202 — 3CEIT202 DATA MINING AND DATA WAREHOUSING

course_id	COURSE-3CEIT202
course_code [O]	3CEIT202
course_name [O]	DATA MINING AND DATA WAREHOUSING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-001 Dr. Meera Datavala
laboratory [S]	LAB-IT-003 Database Systems Lab (Room DA-303)

synthetic description [S]	—

COURSE-3CEIT203 — 3CEIT203 CLOUD COMPUTING

course_id	COURSE-3CEIT203
course_code [O]	3CEIT203
course_name [O]	CLOUD COMPUTING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-009 Dr. Aditi Cloudani
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	—

COURSE-3CEIT204 — 3CEIT204 ADVANCED OPERATING SYSTEM

course_id	COURSE-3CEIT204
course_code [O]	3CEIT204
course_name [O]	ADVANCED OPERATING SYSTEM
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	elective — group EGRP-MTECH-IT-02-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-014 Prof. Rohan Kernelwala
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Elective option in group ELECTIVE-III: a course on Advanced Operating System covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT205 — 3CEIT205 ADVANCED COMPUTER ARCHITECTURE

course_id	COURSE-3CEIT205
course_code [O]	3CEIT205
course_name [O]	ADVANCED COMPUTER ARCHITECTURE
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	elective — group EGRP-MTECH-IT-02-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-014 Prof. Rohan Kernelwala
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Elective option in group ELECTIVE-III: a course on Advanced Computer Architecture covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT206 — 3CEIT206 SOFTWARE ARCHITECTURE AND DESIGN PATTERN

course_id	COURSE-3CEIT206
course_code [O]	3CEIT206
course_name [O]	SOFTWARE ARCHITECTURE AND DESIGN PATTERN
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	elective — group EGRP-MTECH-IT-02-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-017 Dr. Neha Testani
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Elective option in group ELECTIVE-III: a course on Software Architecture And Design Pattern covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT207 — 3CEIT207 PUBLIC KEY INFRASTRUCTURE

course_id	COURSE-3CEIT207
course_code [O]	3CEIT207
course_name [O]	PUBLIC KEY INFRASTRUCTURE
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	elective — group EGRP-MTECH-IT-02-ELECTIVE-III
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-007 Dr. Sana Cipherwala
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-III: a course on Public Key Infrastructure covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT208 — 3CEIT208 SATELLITE NETWORKS

course_id	COURSE-3CEIT208
course_code [O]	3CEIT208
course_name [O]	SATELLITE NETWORKS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	elective — group EGRP-MTECH-IT-02-ELECTIVE-IV
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-016 Prof. Aman Circuitkar
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-IV: a course on Satellite Networks covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT209 — 3CEIT209 ESSENTIAL MATHEMATICS

course_id	COURSE-3CEIT209
course_code [O]	3CEIT209
course_name [O]	ESSENTIAL MATHEMATICS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	elective — group EGRP-MTECH-IT-02-ELECTIVE-IV
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-019 Dr. Farhan Pythonwala
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Elective option in group ELECTIVE-IV: a course on Essential Mathematics covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT210 — 3CEIT210 MODELING AND SIMULATION

course_id	COURSE-3CEIT210
course_code [O]	3CEIT210
course_name [O]	MODELING AND SIMULATION
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	elective — group EGRP-MTECH-IT-02-ELECTIVE-IV
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-019 Dr. Farhan Pythonwala
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Elective option in group ELECTIVE-IV: a course on Modeling And Simulation covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT211 — 3CEIT211 ADVANCED DATABASE MANAGEMENT SYSTEM

course_id	COURSE-3CEIT211
course_code [O]	3CEIT211
course_name [O]	ADVANCED DATABASE MANAGEMENT SYSTEM
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	elective — group EGRP-MTECH-IT-02-ELECTIVE-IV
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-001 Dr. Meera Datavala
laboratory [S]	LAB-IT-003 Database Systems Lab (Room DA-303)
synthetic description [S]	Elective option in group ELECTIVE-IV: a course on Advanced Database Management System covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT301 — 3CEIT301 MAJOR PROJECT PART - I

course_id	COURSE-3CEIT301
course_code [O]	3CEIT301
course_name [O]	MAJOR PROJECT PART - I
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [O]
course_type	regular
credits	8.0 (TH 0.0 + PR/TW 8.0); 16.0 contact hours; total marks 200 (TH int/ext None/None, PR int/ext 100/100) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—
synthetic description [S]	First part of the M.Tech dissertation: problem formulation, literature review and research proposal.

COURSE-3CEIT302 — 3CEIT302 SEMINAR

course_id	COURSE-3CEIT302
course_code [O]	3CEIT302
course_name [O]	SEMINAR
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [O]
course_type	regular
credits	2.0 (TH 0.0 + PR/TW 2.0); 4.0 contact hours; total marks 100 (TH int/ext None/None, PR int/ext 50/50) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—

synthetic description [S]	—

COURSE-3CEIT303 — 3CEIT303 SEMANTIC WEB AND SOCIAL NETWORKS

course_id	COURSE-3CEIT303
course_code [O]	3CEIT303
course_name [O]	SEMANTIC WEB AND SOCIAL NETWORKS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [DO]
course_type	elective — group EGRP-MTECH-IT-03-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-003 Dr. Nisha Tokenwala
laboratory [S]	LAB-IT-002 Natural Language Processing Lab (Room DA-302)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Semantic Web And Social Networks covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT304 — 3CEIT304 DATA COMPRESSION

course_id	COURSE-3CEIT304
course_code [O]	3CEIT304
course_name [O]	DATA COMPRESSION
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [DO]
course_type	elective — group EGRP-MTECH-IT-03-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-019 Dr. Farhan Pythonwala
laboratory [S]	LAB-IT-004 Programming Lab (Room DA-304)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Data Compression covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT305 — 3CEIT305 INTERNET OF THINGS

course_id	COURSE-3CEIT305
course_code [O]	3CEIT305
course_name [O]	INTERNET OF THINGS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [DO]
course_type	elective — group EGRP-MTECH-IT-03-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-016 Prof. Aman Circuitkar
laboratory [S]	LAB-IT-007 IoT & Robotics Lab (Room DB-101)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Internet Of Things covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT306 — 3CEIT306 BIG DATA ANALYTICS

course_id	COURSE-3CEIT306
course_code [O]	3CEIT306
course_name [O]	BIG DATA ANALYTICS
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [DO]
course_type	elective — group EGRP-MTECH-IT-03-ELECTIVE-V
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-010 Dr. Harsh Clusterwala
laboratory [S]	LAB-IT-006 Cloud & Big Data Lab (Room DA-402)
synthetic description [S]	Elective option in group ELECTIVE-V: a course on Big Data Analytics covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT307 — 3CEIT307 NATURAL LANGUAGE PROCESSING

course_id	COURSE-3CEIT307
course_code [O]	3CEIT307
course_name [O]	NATURAL LANGUAGE PROCESSING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [DO]
course_type	elective — group EGRP-MTECH-IT-03-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-003 Dr. Nisha Tokenwala
laboratory [S]	LAB-IT-002 Natural Language Processing Lab (Room DA-302)

synthetic description [S]	Advanced NLP: neural language models, information extraction, question answering and evaluation.

COURSE-3CEIT308 — 3CEIT308 COMPILER DESIGN

course_id	COURSE-3CEIT308
course_code [O]	3CEIT308
course_name [O]	COMPILER DESIGN
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [DO]
course_type	elective — group EGRP-MTECH-IT-03-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-013 Dr. Pooja Lambdani
laboratory [S]	LAB-IT-010 Systems & Software Engineering Lab (Room DA-305)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on Compiler Design covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT309 — 3CEIT309 DEEP LEARNING

course_id	COURSE-3CEIT309
course_code [O]	3CEIT309
course_name [O]	DEEP LEARNING
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [DO]
course_type	elective — group EGRP-MTECH-IT-03-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-004 Dr. Kabir Modelkar
laboratory [S]	LAB-IT-001 AI & Machine Learning Lab (Room DA-301)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on Deep Learning covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT310 — 3CEIT310 CYBER SECURITY

course_id	COURSE-3CEIT310
course_code [O]	3CEIT310
course_name [O]	CYBER SECURITY
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-03 (MTECH IT Semester III) [DO]
course_type	elective — group EGRP-MTECH-IT-03-ELECTIVE-VI
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.107 — M.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	FAC-IT-007 Dr. Sana Cipherwala
laboratory [S]	LAB-IT-005 Networks & Security Lab (Room DA-401)
synthetic description [S]	Elective option in group ELECTIVE-VI: a course on Cyber Security covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-3CEIT401 — 3CEIT401 MAJOR PROJECT PART - II

course_id	COURSE-3CEIT401
course_code [O]	3CEIT401
course_name [O]	MAJOR PROJECT PART - II
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-04 (MTECH IT Semester IV) [O]
course_type	regular
credits	12.0 (TH 0.0 + PR/TW 12.0); 24.0 contact hours; total marks 300 (TH int/ext None/None, PR int/ext 150/150) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.108 — M.TECH-IT Semester IV – course table
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	COURSE-3CEIT301 MAJOR PROJECT PART - I
faculty [S]	FAC-IT-020 Dr. Leena Projectkar
laboratory [S]	—
synthetic description [S]	Second part of the M.Tech dissertation: implementation, experiments, thesis and publication.

C. Common / basic courses in the B.Tech IT and M.Tech IT schemes

COURSE-2AC103 — 2AC103 ENVIRONMENTAL STUDIES AND DISASTER MANAGEMENT

course_id	COURSE-2AC103
course_code [O]	2AC103
course_name [O]	ENVIRONMENTAL STUDIES AND DISASTER MANAGEMENT
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-01 (BTECH IT Semester I), SEM-BTECH-IT-02 (BTECH IT Semester II) [DO]
course_type	audit — group EGRP-BTECH-IT-01-AUDIT
credits	0.0 (TH 0.0 + PR/TW 0.0); 2.0 contact hours; total marks 0 (TH int/ext None/None, PR int/ext None/None) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.10 — B.TECH-IT Semester I – List of Audit Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.TECH-CE(AI), B.TECH-ICT
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 216 BTech(Automobile Engineering) (p.39); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2BS101 — 2BS101 MATHEMATICS - I

course_id	COURSE-2BS101
course_code [O]	2BS101
course_name [O]	MATHEMATICS - I
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-01 (BTECH IT Semester I) [O]
course_type	regular
credits	4.0 (TH 4.0 + PR/TW 0.0); 4.0 contact hours; total marks 100 (TH int/ext 40/60, PR int/ext None/None) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.10 — B.TECH-IT Semester I – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 246 BSc(Nautical Science) (p.39); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2ES101 — 2ES101 ENGINEERING GRAPHICS

course_id	COURSE-2ES101
course_code [O]	2ES101
course_name [O]	ENGINEERING GRAPHICS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-01 (BTECH IT Semester I) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.10 — B.TECH-IT Semester I – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 216 BTech(Automobile Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2ES103 — 2ES103 BASIC ELECTRICAL ENGINEERING

course_id	COURSE-2ES103
course_code [O]	2ES103
course_name [O]	BASIC ELECTRICAL ENGINEERING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-01 (BTECH IT Semester I) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.10 — B.TECH-IT Semester I – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2018-2019 under code 203 BTech(Mechanical Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2019-2020 under code 203 BTech(Mechanical Engineering) (p.62); 2020-2021 under code 216 BTech(Automobile Engineering) (p.40); 2020-2021 under code 246 BSc(Nautical Science) (p.39); 2021-2022 under code 201 BTech(Computer Engineering) (p.2); 2021-2022 under code 203 BTech(Mechanical Engineering) (p.3)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2ES109 — 2ES109 BASICS OF WEB TECHNOLOGY

course_id	COURSE-2ES109
course_code [O]	2ES109
course_name [O]	BASICS OF WEB TECHNOLOGY
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-01 (BTECH IT Semester I) [O]
course_type	regular
credits	4.0 (TH 3.0 + PR/TW 1.0); 5.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.10 — B.TECH-IT Semester I – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2ES110 — 2ES110 IT PERIPHERALS & TOOLS

course_id	COURSE-2ES110
course_code [O]	2ES110
course_name [O]	IT PERIPHERALS & TOOLS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-01 (BTECH IT Semester I) [O]
course_type	regular
credits	2.0 (TH 0.0 + PR/TW 2.0); 4.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.10 — B.TECH-IT Semester I – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2HS102 — 2HS102 SEMINAR

course_id	COURSE-2HS102
course_code [O]	2HS102
course_name [O]	SEMINAR
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-01 (BTECH IT Semester I) [O]
course_type	regular
credits	2.0 (TH 0.0 + PR/TW 2.0); 4.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.10 — B.TECH-IT Semester I – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-BME
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 205 BTech(Biomedical Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—

synthetic description [S]	—

COURSE-2BS102 — 2BS102 MATHEMATICS - II

course_id	COURSE-2BS102
course_code [O]	2BS102
course_name [O]	MATHEMATICS - II
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-02 (BTECH IT Semester II) [O]
course_type	regular
credits	4.0 (TH 4.0 + PR/TW 0.0); 4.0 contact hours; total marks 100 (TH int/ext 40/60, PR int/ext None/None) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.11 — B.TECH-IT Semester II – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 246 BSc(Nautical Science) (p.39); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2BS103 — 2BS103 PHYSICS

course_id	COURSE-2BS103
course_code [O]	2BS103
course_name [O]	PHYSICS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-02 (BTECH IT Semester II) [O]
course_type	regular
credits	3.0 (TH 2.0 + PR/TW 1.0); 4.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.11 — B.TECH-IT Semester II – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2018-2019 under code 203 BTech(Mechanical Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2019-2020 under code 203 BTech(Mechanical Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.40); 2020-2021 under code 246 BSc(Nautical Science) (p.39); 2021-2022 under code 201 BTech(Computer Engineering) (p.2); 2021-2022 under code 203 BTech(Mechanical Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2ES102 — 2ES102 WORKSHOP MANUFACTURING PRACTICE

course_id	COURSE-2ES102
course_code [O]	2ES102
course_name [O]	WORKSHOP MANUFACTURING PRACTICE
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-02 (BTECH IT Semester II) [O]
course_type	regular
credits	2.0 (TH 0.0 + PR/TW 2.0); 4.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.11 — B.TECH-IT Semester II – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2018-2019 under code 203 BTech(Mechanical Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2019-2020 under code 203 BTech(Mechanical Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2020-2021 under code 216 BTech(Automobile Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2); 2021-2022 under code 203 BTech(Mechanical Engineering) (p.3)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2ES104 — 2ES104 PROGRAMMING FOR PROBLEM SOLVING

course_id	COURSE-2ES104
course_code [O]	2ES104
course_name [O]	PROGRAMMING FOR PROBLEM SOLVING
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-02 (BTECH IT Semester II) [O]
course_type	regular
credits	4.0 (TH 2.0 + PR/TW 2.0); 6.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.11 — B.TECH-IT Semester II – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 216 BTech(Automobile Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2ES111 — 2ES111 IT WORKSHOP

course_id	COURSE-2ES111
course_code [O]	2ES111
course_name [O]	IT WORKSHOP
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-02 (BTECH IT Semester II) [O]
course_type	regular
credits	2.0 (TH 0.0 + PR/TW 2.0); 4.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.11 — B.TECH-IT Semester II – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2HS101 — 2HS101 COMMUNICATION SKILLS

course_id	COURSE-2HS101
course_code [O]	2HS101
course_name [O]	COMMUNICATION SKILLS
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-02 (BTECH IT Semester II) [O]
course_type	regular
credits	3.0 (TH 2.0 + PR/TW 1.0); 4.0 contact hours; total marks 150 (TH int/ext 40/60, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.11 — B.TECH-IT Semester II – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI)
employability catalog [O]	2018-2019 under code 201 BTech(Computer Engineering) (p.89); 2018-2019 under code 203 BTech(Mechanical Engineering) (p.89); 2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2019-2020 under code 203 BTech(Mechanical Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.41); 2020-2021 under code 246 BSc(Nautical Science) (p.39); 2021-2022 under code 201 BTech(Computer Engineering) (p.2); 2021-2022 under code 203 BTech(Mechanical Engineering) (p.3)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-UHV1IPDC1 — UHV1IPDC1 UNIVERSAL HUMAN VALUE (INTEGRATED PERSONALITY DEVELOPMENT COURSE-1)

course_id	COURSE-UHV1IPDC1
course_code [O]	UHV1IPDC1
course_name [O]	UNIVERSAL HUMAN VALUE (INTEGRATED PERSONALITY DEVELOPMENT COURSE-1)
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-02 (BTECH IT Semester II) [DO]
course_type	elective — group EGRP-BTECH-IT-02-UHV-ELECTIVE-I
credits	2.0 (TH 2.0 + PR/TW 0.0); 2.0 contact hours; total marks 100 (TH int/ext 100/None, PR int/ext None/None) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.11 — B.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.TECH-CE(AI), B.TECH-CSBS, B.TECH-CSE(BDA), B.TECH-CSE(CBA), B.TECH-CSE(CS), B.TECH-ICT
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	Elective option in group UHV-ELECTIVE-I: a course on Universal Human Value (Integrated Personality Development Course-1) covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-UHV3DYSBG1 — UHV3DYSBG1 UNIVERSAL HUMAN VALUE (DISCOVER YOURSELF THROUGH BHAGAVAD GITA-1)

course_id	COURSE-UHV3DYSBG1
course_code [O]	UHV3DYSBG1
course_name [O]	UNIVERSAL HUMAN VALUE (DISCOVER YOURSELF THROUGH BHAGAVAD GITA-1)
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-02 (BTECH IT Semester II) [DO]
course_type	elective — group EGRP-BTECH-IT-02-UHV-ELECTIVE-I
credits	2.0 (TH 2.0 + PR/TW 0.0); 2.0 contact hours; total marks 100 (TH int/ext 100/None, PR int/ext None/None) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.11 — B.TECH-IT Semester II – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI), B.TECH-CSBS, B.TECH-CSE(BDA), B.TECH-CSE(CBA), B.TECH-CSE(CS), B.TECH-ICT
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	Elective option in group UHV-ELECTIVE-I: a course on Universal Human Value (Discover Yourself Through Bhagavad Gita-1) covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2AC301 — 2AC301 AUDIT COURSE

course_id	COURSE-2AC301
course_code [O]	2AC301
course_name [O]	AUDIT COURSE
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-03 (BTECH IT Semester III) [DO]
course_type	audit — group EGRP-BTECH-IT-03-AUDIT
credits	0.0 (TH 0.0 + PR/TW 0.0); 2.0 contact hours; total marks 0 (TH int/ext None/None, PR int/ext None/None) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.12 — B.TECH-IT Semester III – List of Audit Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 205 BTech(Biomedical Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2BS3102 — 2BS3102 DISCRETE MATHEMATICS & PROBABILITY

course_id	COURSE-2BS3102
course_code [O]	2BS3102
course_name [O]	DISCRETE MATHEMATICS & PROBABILITY
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-03 (BTECH IT Semester III) [O]
course_type	regular
credits	4.0 (TH 4.0 + PR/TW 0.0); 4.0 contact hours; total marks 100 (TH int/ext 40/60, PR int/ext None/None) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.12 — B.TECH-IT Semester III – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-UHV2IPDC2 — UHV2IPDC2 UNIVERSAL HUMAN VALUE (INTEGRATED PERSONALITY DEVELOPMENT COURSE-2)

course_id	COURSE-UHV2IPDC2
course_code [O]	UHV2IPDC2
course_name [O]	UNIVERSAL HUMAN VALUE (INTEGRATED PERSONALITY DEVELOPMENT COURSE-2)
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-03 (BTECH IT Semester III) [DO]
course_type	elective — group EGRP-BTECH-IT-03-UHV-ELECTIVE-II
credits	2.0 (TH 2.0 + PR/TW 0.0); 2.0 contact hours; total marks 100 (TH int/ext 100/None, PR int/ext None/None) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.12 — B.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI), B.TECH-CSE(BDA), B.TECH-CSE(CBA), B.TECH-CSE(CS), B.TECH-ICT
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	Elective option in group UHV-ELECTIVE-II: a course on Universal Human Value (Integrated Personality Development Course-2) covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-UHV4DYSBG2 — UHV4DYSBG2 UNIVERSAL HUMAN VALUE (DISCOVER YOURSELF THROUGH BHAGAVAD GITA-2)

course_id	COURSE-UHV4DYSBG2
course_code [O]	UHV4DYSBG2
course_name [O]	UNIVERSAL HUMAN VALUE (DISCOVER YOURSELF THROUGH BHAGAVAD GITA-2)
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-03 (BTECH IT Semester III) [DO]
course_type	elective — group EGRP-BTECH-IT-03-UHV-ELECTIVE-II
credits	2.0 (TH 2.0 + PR/TW 0.0); 2.0 contact hours; total marks 100 (TH int/ext 100/None, PR int/ext None/None) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.12 — B.TECH-IT Semester III – List of Elective Subjects
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-MR, B.TECH-AUTO, B.TECH-PETRO, B.SC.-NAUTICAL, B.TECH-CE(AI), B.TECH-CSE(BDA), B.TECH-CSE(CBA), B.TECH-CSE(CS), B.TECH-ICT
employability catalog [O]	2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	Elective option in group UHV-ELECTIVE-II: a course on Universal Human Value (Discover Yourself Through Bhagavad Gita-2) covering its core concepts, tools and a mini project. (Generic synthetic summary.)

COURSE-2BS4101 — 2BS4101 MATHEMATICS FOR COMPUTER ENGINEERING & INFORMATION TECHNOLOGY

course_id	COURSE-2BS4101
course_code [O]	2BS4101
course_name [O]	MATHEMATICS FOR COMPUTER ENGINEERING & INFORMATION TECHNOLOGY
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-04 (BTECH IT Semester IV) [O]
course_type	regular
credits	4.0 (TH 4.0 + PR/TW 0.0); 4.0 contact hours; total marks 100 (TH int/ext 40/60, PR int/ext None/None) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.13 — B.TECH-IT Semester IV – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-CE(AI)
employability catalog [O]	2019-2020 under code 201 BTech(Computer Engineering) (p.62); 2020-2021 under code 201 BTech(Computer Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2HS5101 — 2HS5101 APTITUDE SKILL BUILDING-I

course_id	COURSE-2HS5101
course_code [O]	2HS5101
course_name [O]	APTITUDE SKILL BUILDING-I
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-05 (BTECH IT Semester V) [O]
course_type	regular
credits	1.0 (TH 0.0 + PR/TW 1.0); 2.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.14 — B.TECH-IT Semester V – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-PETRO, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 205 BTech(Biomedical Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-2HS6101 — 2HS6101 APTITUDE SKILL BUILDING–II

course_id	COURSE-2HS6101
course_code [O]	2HS6101
course_name [O]	APTITUDE SKILL BUILDING–II
program [O]	PROG-BTECH-IT-001
semester	SEM-BTECH-IT-06 (BTECH IT Semester VI) [O]
course_type	regular
credits	1.0 (TH 0.0 + PR/TW 1.0); 2.0 contact hours; total marks 50 (TH int/ext None/None, PR int/ext 30/20) [O]
official source	DOC-OFFICIAL-SYLLABUS-001 p.15 — B.TECH-IT Semester VI – course table
also in syllabus of [DO]	B.TECH-CE, B.TECH-ME, B.TECH-MC, B.TECH-BME, B.TECH-CIVIL, B.TECH-EE, B.TECH-PETRO, B.TECH-CE(AI)
employability catalog [O]	2020-2021 under code 205 BTech(Biomedical Engineering) (p.40); 2021-2022 under code 201 BTech(Computer Engineering) (p.2)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

COURSE-3OS201 — 3OS201 OPEN SUBJECT

course_id	COURSE-3OS201
course_code [O]	3OS201
course_name [O]	OPEN SUBJECT
program [O]	PROG-MTECH-IT-001
semester	SEM-MTECH-IT-02 (MTECH IT Semester II) [DO]
course_type	open — group EGRP-MTECH-IT-02-OPEN
credits	3.0 (TH 3.0 + PR/TW 0.0); 3.0 contact hours; total marks 100 (TH int/ext 40/60, PR int/ext None/None) [DO: inherited from slot]
official source	DOC-OFFICIAL-SYLLABUS-001 p.105 — M.TECH-IT Semester II – List of Open Subjects
also in syllabus of [DO]	M.TECH-CE
employability catalog [O]	2019-2020 under code 401 Mtech(Computer Engineering) (p.77); 2020-2021 under code 401 Mtech(Computer Engineering) (p.55); 2021-2022 under code 401 Mtech(Computer Engineering) (p.20)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	Not assigned in demo dataset
laboratory [S]	—
synthetic description [S]	—

D. Legacy IT course codes found only in the certified course list

COURSE-2IT302 — 2IT302 COMPUTER SYSTEM ORGANIZATION

course_id	COURSE-2IT302
course_code [O]	2IT302
course_name [O]	COMPUTER SYSTEM ORGANIZATION
program [O]	PROG-BTECH-IT-001
semester	Not available (not in 2021-2022 syllabus) [DO]
course_type	not available
credits	Not available
official source	DOC-OFFICIAL-CATALOG-001 p.112 — Year-wise course list
also in syllabus of [DO]	—
employability catalog [O]	2017-2018 under code 202 BTech(Information Technology) (p.112); 2018-2019 under code 202 BTech(Information Technology) (p.89)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	—
laboratory [S]	—
synthetic description [S]	—
curriculum note [DO]	Listed under catalog program code 202 (BTech(Information Technology)) in 2017-2018, 2018-2019; not present in the 2021-2022 structured syllabus for this program. Semester, credits and scheme are unavailable.

COURSE-2IT503 — 2IT503 COMPUTER SYSTEM ARCHITECTURE

course_id	COURSE-2IT503
course_code [O]	2IT503
course_name [O]	COMPUTER SYSTEM ARCHITECTURE
program [O]	PROG-BTECH-IT-001
semester	Not available (not in 2021-2022 syllabus) [DO]
course_type	not available
credits	Not available
official source	DOC-OFFICIAL-CATALOG-001 p.112 — Year-wise course list
also in syllabus of [DO]	—
employability catalog [O]	2017-2018 under code 202 BTech(Information Technology) (p.112); 2018-2019 under code 202 BTech(Information Technology) (p.89); 2019-2020 under code 202 BTech(Information Technology) (p.62)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	—
laboratory [S]	—
synthetic description [S]	—
curriculum note [DO]	Listed under catalog program code 202 (BTech(Information Technology)) in 2017-2018, 2018-2019, 2019-2020; not present in the 2021-2022 structured syllabus for this program. Semester, credits and scheme are unavailable.

COURSE-2IT606 — 2IT606 IT INDUSTRY MANAGEMENT

course_id	COURSE-2IT606
course_code [O]	2IT606
course_name [O]	IT INDUSTRY MANAGEMENT
program [O]	PROG-BTECH-IT-001
semester	Not available (not in 2021-2022 syllabus) [DO]
course_type	not available
credits	Not available
official source	DOC-OFFICIAL-CATALOG-001 p.112 — Year-wise course list
also in syllabus of [DO]	—
employability catalog [O]	2017-2018 under code 202 BTech(Information Technology) (p.112); 2018-2019 under code 202 BTech(Information Technology) (p.89); 2019-2020 under code 202 BTech(Information Technology) (p.62)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	—
laboratory [S]	—
synthetic description [S]	—
curriculum note [DO]	Listed under catalog program code 202 (BTech(Information Technology)) in 2017-2018, 2018-2019, 2019-2020; not present in the 2021-2022 structured syllabus for this program. Semester, credits and scheme are unavailable.

COURSE-3IT101 — 3IT101 SOFT COMPUTING

course_id	COURSE-3IT101
course_code [O]	3IT101
course_name [O]	SOFT COMPUTING
program [O]	PROG-MTECH-IT-001
semester	Not available (not in 2021-2022 syllabus) [DO]
course_type	not available
credits	Not available
official source	DOC-OFFICIAL-CATALOG-001 p.129 — Year-wise course list
also in syllabus of [DO]	—
employability catalog [O]	2017-2018 under code 402 Mtech(Information Technology) (p.129); 2018-2019 under code 402 Mtech(Information Technology) (p.102)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	—
laboratory [S]	—
synthetic description [S]	—
curriculum note [DO]	Listed under catalog program code 402 (Mtech(Information Technology)) in 2017-2018, 2018-2019; not present in the 2021-2022 structured syllabus for this program. Semester, credits and scheme are unavailable.

COURSE-3IT201 — 3IT201 FUNDAMENTALS OF IMAGE PROCESSING

course_id	COURSE-3IT201
course_code [O]	3IT201
course_name [O]	FUNDAMENTALS OF IMAGE PROCESSING
program [O]	PROG-MTECH-IT-001
semester	Not available (not in 2021-2022 syllabus) [DO]
course_type	not available
credits	Not available
official source	DOC-OFFICIAL-CATALOG-001 p.129 — Year-wise course list
also in syllabus of [DO]	—
employability catalog [O]	2017-2018 under code 402 Mtech(Information Technology) (p.129); 2018-2019 under code 402 Mtech(Information Technology) (p.102)
prerequisites (official)	Not specified in official sources
synthetic prerequisites [S]	—
faculty [S]	—
laboratory [S]	—
synthetic description [S]	—
curriculum note [DO]	Listed under catalog program code 402 (Mtech(Information Technology)) in 2017-2018, 2018-2019; not present in the 2021-2022 structured syllabus for this program. Semester, credits and scheme are unavailable.

E. Synthetic advisory prerequisite map (REQUIRES)

source_type: **synthetic (CampusMate demo data)** · not an official GUNI prerequisite list

Prerequisite ID	Course	Requires	Strength
PREREQ-SYN-001	2CEIT302 OBJECT ORIENTED PROGRAMMING	2ES104 PROGRAMMING FOR PROBLEM SOLVING	recommended
PREREQ-SYN-002	2CEIT303 DATABASE MANAGEMENT SYSTEM	2ES104 PROGRAMMING FOR PROBLEM SOLVING	recommended
PREREQ-SYN-003	2CEIT304 DATA STRUCTURES	2ES104 PROGRAMMING FOR PROBLEM SOLVING	recommended
PREREQ-SYN-004	2CEIT401 OPERATING SYSTEMS	2CEIT304 DATA STRUCTURES	recommended
PREREQ-SYN-005	2CEIT402 DESIGN AND ANALYSIS OF ALGORITHMS	2CEIT304 DATA STRUCTURES	recommended
PREREQ-SYN-006	2CEIT405 NOSQL DATABASE SYSTEMS	2CEIT303 DATABASE MANAGEMENT SYSTEM	recommended
PREREQ-SYN-007	2CEIT501 COMPUTER ARCHITECTURE & ORGANIZATION	2CEIT301 DIGITAL ELECTRONICS	recommended
PREREQ-SYN-008	2CEIT503 COMPUTER NETWORKS	2CEIT401 OPERATING SYSTEMS	recommended
PREREQ-SYN-009	2CEIT601 THEORY OF COMPUTATION	2BS3102 DISCRETE MATHEMATICS & PROBABILITY	recommended
PREREQ-SYN-010	2CEIT602 ARTIFICIAL INTELLIGENCE	2CEIT402 DESIGN AND ANALYSIS OF ALGORITHMS	recommended
PREREQ-SYN-011	2CEIT602 ARTIFICIAL INTELLIGENCE	2BS3102 DISCRETE MATHEMATICS & PROBABILITY	recommended
PREREQ-SYN-012	2CEIT603 CLOUD COMPUTING	2CEIT503 COMPUTER NETWORKS	recommended
PREREQ-SYN-013	2CEIT701 COMPILER DESIGN	2CEIT601 THEORY OF COMPUTATION	recommended
PREREQ-SYN-014	2CEIT702 BIG DATA ANALYTICS	2CEIT303 DATABASE MANAGEMENT SYSTEM	recommended
PREREQ-SYN-015	2CEIT702 BIG DATA ANALYTICS	2CEIT405 NOSQL DATABASE SYSTEMS	recommended
PREREQ-SYN-016	2CEIT6PE4 DATA SCIENCES AND MACHINE LEARNING	2CEIT404 PROGRAMMING WITH PYTHON	recommended
PREREQ-SYN-017	2CEIT6PE3 CRYPTOGRAPHY AND NETWORK SECURITY	2CEIT503 COMPUTER NETWORKS	recommended
PREREQ-SYN-018	2CEIT6PE7 ETHICAL HACKING	2CEIT503 COMPUTER NETWORKS	recommended
PREREQ-SYN-019	2CEIT78PE1 DEEP LEARNING	2CEIT602 ARTIFICIAL INTELLIGENCE	recommended
PREREQ-SYN-020	2CEIT78PE7 NATURAL LANGUAGE PROCESSING	2CEIT602 ARTIFICIAL INTELLIGENCE	recommended
PREREQ-SYN-021	2CEIT78PE13 COMPUTER VISION	2CEIT602 ARTIFICIAL INTELLIGENCE	recommended
PREREQ-SYN-022	2CEIT605 CAPSTONE PROJECT-II	2CEIT504 CAPSTONE PROJECT-I	recommended
PREREQ-SYN-023	2CEIT703 CAPSTONE PROJECT-III	2CEIT605 CAPSTONE PROJECT-II	recommended
PREREQ-SYN-024	3CEIT401 MAJOR PROJECT PART - II	3CEIT301 MAJOR PROJECT PART - I	recommended

F. Data-quality findings in the official sources

source_type: **derived_official (structured from GUNI PDF)**

Conflict ID	Code	Name in syllabus	Name in catalog	Year / program code	Catalog page
DQ-CONFLICT-001	2HS101	COMMUNICATION SKILLS	CALCULUS	2017-2018 / 201	112
DQ-CONFLICT-002	2HS101	COMMUNICATION SKILLS	CALCULUS	2017-2018 / 205	114
DQ-CONFLICT-003	2HS101	COMMUNICATION SKILLS	CALCULUS	2019-2020 / 215	68
DQ-CONFLICT-004	2HS101	COMMUNICATION SKILLS	CALCULUS	2020-2021 / 215	45

Conflict ID	Code	Name in syllabus	Name in catalog	Year / program code	Catalog page
DQ-CONFLICT-005	2HS102	SEMINAR	LINEAR ALGEBRA	2017-2018 / 201	112
DQ-CONFLICT-006	2HS102	SEMINAR	LINEAR ALGEBRA	2017-2018 / 205	114
DQ-CONFLICT-007	2HS102	SEMINAR	LINEAR ALGEBRA	2019-2020 / 215	68
DQ-CONFLICT-008	2HS102	SEMINAR	LINEAR ALGEBRA	2020-2021 / 215	45
DQ-CONFLICT-009	3OS201	OPEN SUBJECT	ENGLISH LANGUAGE & COMMUNICATION SKILLS	2017-2018 / 401	129
DQ-CONFLICT-010	3OS201	OPEN SUBJECT	ENGLISH LANGUAGE & COMMUNICATION SKILLS	2018-2019 / 401	102